Lecture 4 – Regular expressions, Kleene's theorem, string substitution

NTIN071 Automata and Grammars

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Spring 2026

** Adapted from the Czech-lecture slides by Marta Vomlelová with gratitude. The translation, some modifications, and all errors are mine.*

Recap of Lecture 3

- Nondeterministic finite automata (NFA): can 'guess' the right path to accepting, computation described by a state tree.
- ϵ-transitions: allow to change states without reading any input
- Subset construction: every NFA and ϵ-NFA is equivalent to a DFA (but can be easier to design, much smaller).
- Regular languages are closed under set operations (union, intersection, complement, difference)
- And under string operations (concatenation, iteration and positive iteration, reverse, left and right quotient)

1.8 Regular expressions

Regular expressions (RE)

- an algebraic description of languages
- declarative: express the form of the words we want to accept
- can describe all, and only, regular languages
- can be viewed as a programming language, a user/friendly description of a finite automaton

Example

- `grep` command in UNIX.
- Python module `re`
- lexical analysis, e.g. `Flex` (description via 'tokens' $\leftrightsquigarrow$ RE)

Note: syntax analysis needs a stronger tool, context-free grammars

The definition

A regular expression α over (finite, nonempty) Σ, $\alpha \in \mathrm{RegE}(\Sigma)$ and the matching language $L(\alpha)$, are defined inductively:

expression	**language**	**note**
$\emptyset$	$L(\emptyset) = \emptyset$	empty expression
ϵ	$L(\epsilon) = \{\epsilon\}$	empty string
$\mathbf{a}$	$L(\mathbf{a}) = \{a\}$	for all $a \in \Sigma$
$(\alpha + \beta)$	$L((\alpha + \beta)) = L(\alpha) \cup L(\beta)$	union (`grep`, `re` use '|')
$(\alpha\beta)$	$L((\alpha\beta)) = L(\alpha)L(\beta)$	concatenation
α^*	$L(\alpha^*) = L(\alpha)^*$	iteration (Kleene star)

Examples, notation

Example

- The language of alternating 0s and 1s can be expressed as:
  - $(\mathbf{01})^* + (\mathbf{10})^* + \mathbf{1}(\mathbf{01})^* + \mathbf{0}(\mathbf{10})^*$
  - $(\epsilon + \mathbf{1})(\mathbf{01})^*(\epsilon + \mathbf{0})$
- $L((\mathbf{0}^*\mathbf{10}^*\mathbf{10}^*\mathbf{1})^*\mathbf{0}^*) = \{w \in \{0,1\}^* \mid |w|_1 \equiv 0 \pmod 3\}$

We often omit parentheses:

- priority of operators: iteration $*$ > concatenation > union $+$
- associativity of concatenation, union $+$
- outer parentheses

We could define, and will sometimes use, positive iteration α^+

Kleene's theorem

Theorem (Kleene's theorem)

A language is regular, iff it is matched by some regular expression.

We will prove it by giving two constructions:

1. from RE to ϵ-NFA (which can be converted to a DFA)
2. from a DFA to a RE (but we could start from a ϵ-NFA)

For 2. we also mention a better algorithm: state eliminiation

RE to ϵ-NFA

By induction on the structure of α, construct a ϵ-NFA E s.t. $L(\alpha) = L(E)$ with three additional properties:

1. Exactly one accepting state.
2. No incoming edges into the initial state.
3. No outgoing edges from the accepting state.

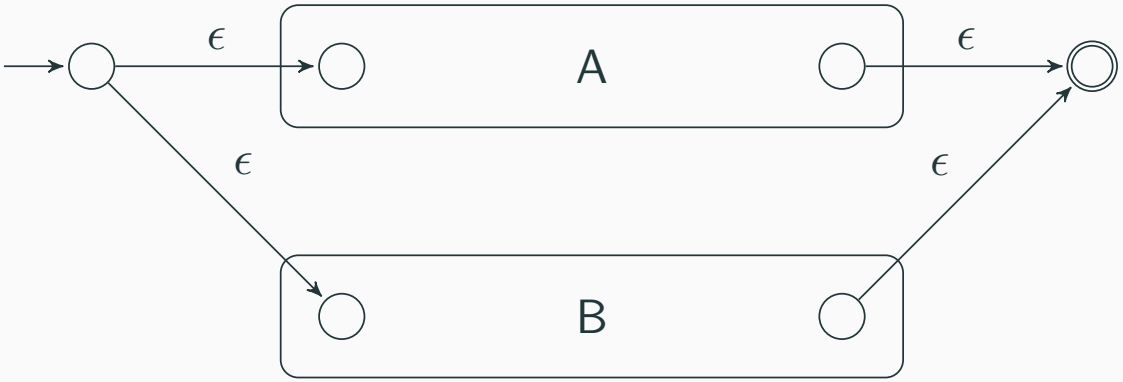
Induction base: α is the empty string ϵ, empty set $\emptyset$, or a letter **a**

Induction step: $\alpha + \beta$, $\alpha\beta$, α^* (next slide) $\square$

RE to ϵ-NFA: Induction step

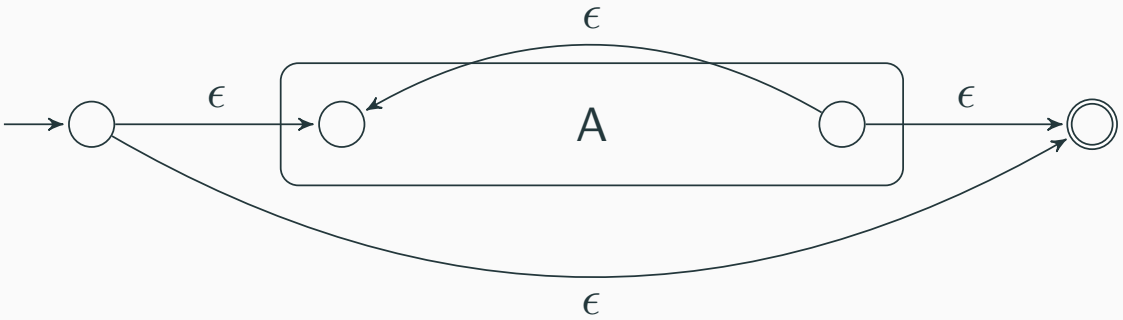
Let A, B be ϵ-NFA constructed for α, β.

Addition $\alpha + \beta$



Concatenation $\alpha\beta$

Iteration α^*



DFA to RE

Assume the states are $Q = \{1, \ldots, n\}$ and the start state is $q_0 = 1$.

Construct a RE $R_{ij}^{(k)}$ matching words that transition from state i into state j and all intermediate states (if any) have index $\leq k$.

Then we set $\alpha = \sum_{j \in F_A} R_{1j}^{(n)}$ (from start to some accepting state)

Iteratively construct $R_{ij}^{(k)}$ for $k = 0, \ldots, n$ (finite induction).

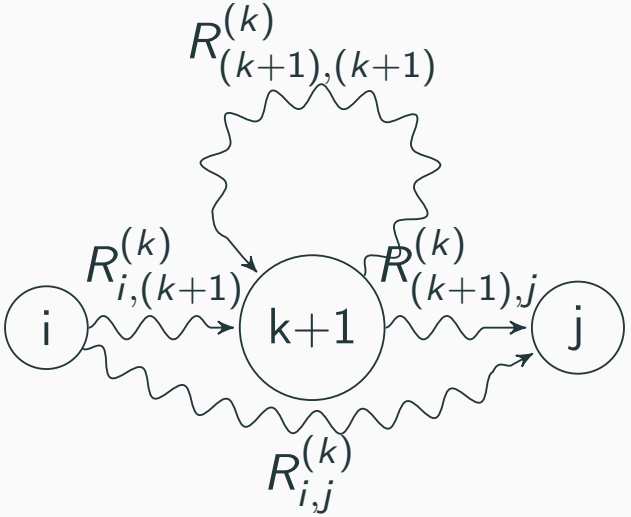
Induction base: $k = 0$

- If $i \neq j$, set $R_{ij}^{(0)} = \mathbf{a_1} + \ldots + \mathbf{a_m}$ where $a_1, \ldots, a_m$ are symbols on edges from i into j ($R_{ij}^{(0)} = \emptyset$ or $R_{ij}^{(0)} = \mathbf{a}$ for $m = 0, 1$).
- If $i = j$, $R_{ii}^{(0)} = \epsilon + \mathbf{a_1} + \ldots + \mathbf{a_m}$ where a_i's are on loops on i.

DFA to RE: Induction step

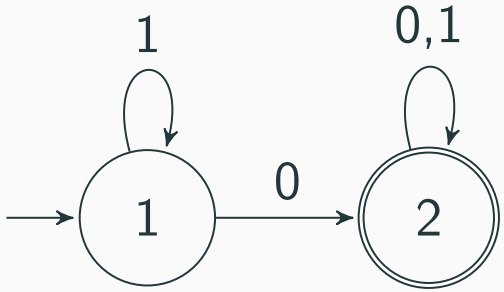
Once we have $R_{ij}^{(k)}$ for all $i, j \in Q$, we can construct $R_{ij}^{(k+1)}$:

$$R_{ij}^{(k+1)} = R_{ij}^{(k)} + R_{i(k+1)}^{(k)} (R_{(k+1)(k+1)}^{(k)})^* R_{(k+1)j}^{(k)}$$



- paths $i \rightsquigarrow j$ not going through $k+1$: already in $R_{ij}^{(k)}$
- paths $i \rightsquigarrow j$ going through $k+1$ one or more times: $i \rightsquigarrow k+1$ (first visit), loop on $k+1$, finally (last visit) $k+1 \rightsquigarrow j$ $\square$

Example



Apply the construction, simplify:

$$\alpha = R_{12}^{(2)} = \mathbf{1}^*\mathbf{0}(\mathbf{0}+\mathbf{1})^*$$

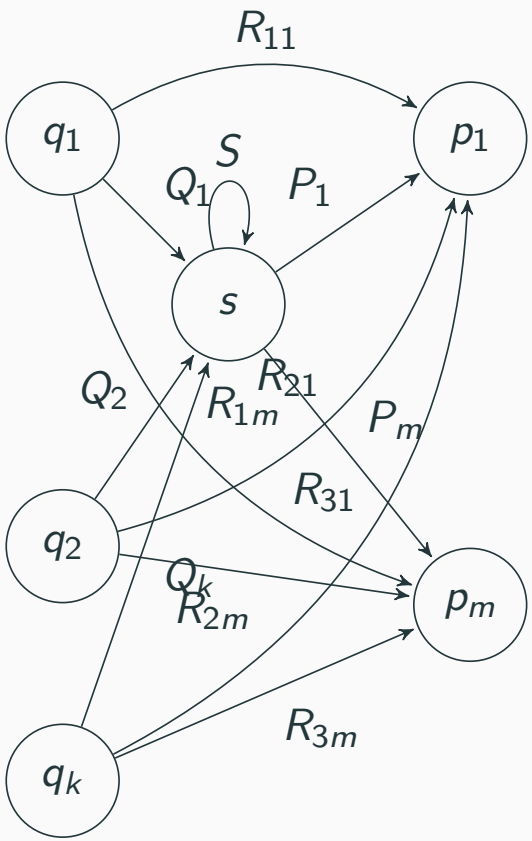
$R_{11}^{(0)}$	$\epsilon + \mathbf{1}$	$=$
$R_{12}^{(0)}$	$\mathbf{0}$	$=$
$R_{21}^{(0)}$	$\emptyset$	$=$
$R_{22}^{(0)}$	$(\epsilon + \mathbf{0} + \mathbf{1})$	$=$
$R_{11}^{(1)}$	$\epsilon + \mathbf{1} + (\epsilon + \mathbf{1})(\epsilon + \mathbf{1})^*(\epsilon + \mathbf{1})$	$=\mathbf{1}^*$
$R_{12}^{(1)}$	$\mathbf{0} + (\epsilon + \mathbf{1})(\epsilon + \mathbf{1})^*\mathbf{0}$	$=\mathbf{1}^*\mathbf{0}$
$R_{21}^{(1)}$	$\emptyset + \emptyset(\epsilon + \mathbf{1})^*(\epsilon + \mathbf{1})$	$=\emptyset$
$R_{22}^{(1)}$	$\epsilon + \mathbf{0} + \mathbf{1} + \emptyset(\epsilon + \mathbf{1})^*\mathbf{0}$	$=\epsilon + \mathbf{0} + \mathbf{1}$
$R_{11}^{(2)}$	$\mathbf{1}^* + \mathbf{1}^*\mathbf{0}(\epsilon + \mathbf{0} + \mathbf{1})^*\emptyset$	$=\mathbf{1}^*$
$R_{12}^{(2)}$	$\mathbf{1}^*\mathbf{0} + \mathbf{1}^*\mathbf{0}(\epsilon + \mathbf{0} + \mathbf{1})^*(\epsilon + \mathbf{0} + \mathbf{1})$	$=\mathbf{1}^*\mathbf{0}(\mathbf{0}+\mathbf{1})^*$
$R_{21}^{(2)}$	$\emptyset + (\epsilon + \mathbf{0} + \mathbf{1})(\epsilon + \mathbf{0} + \mathbf{1})^*\emptyset$	$=\emptyset$
$R_{22}^{(2)}$	$\epsilon + \mathbf{0} + \mathbf{1} + (\epsilon + \mathbf{0} + \mathbf{1})(\epsilon + \mathbf{0} + \mathbf{1})^*(\epsilon + \mathbf{0} + \mathbf{1})$	$=(\mathbf{0}+\mathbf{1})^*$

State elimination algorithm

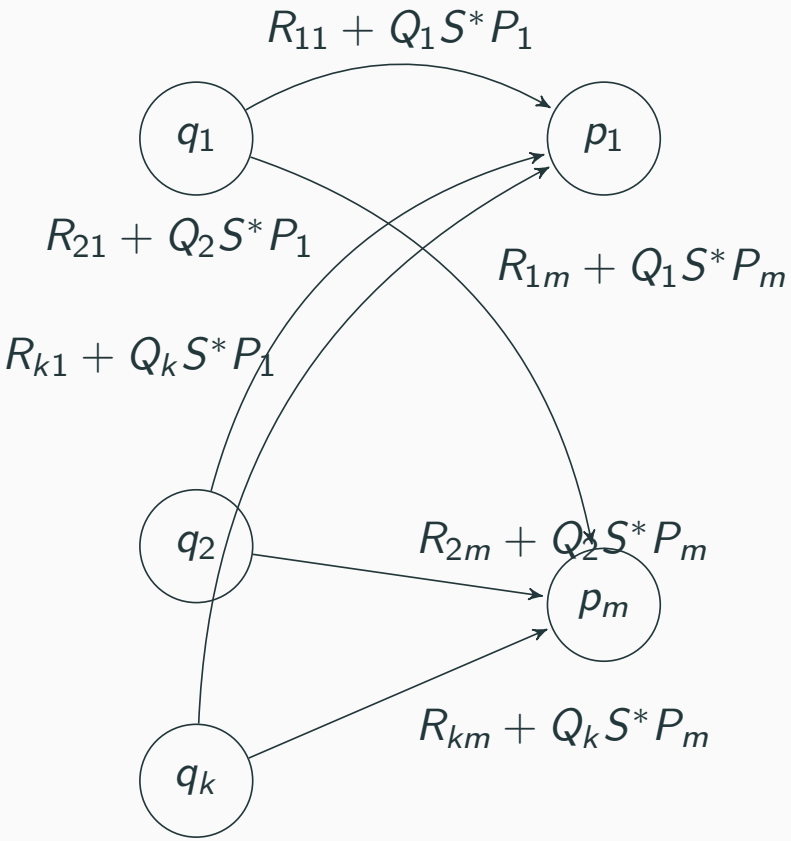
State elimination: the idea

Idea: Allow edges labelled by RE, iteratively remove nodes. (More efficient, avoids duplicity.)

State s selected for elimination



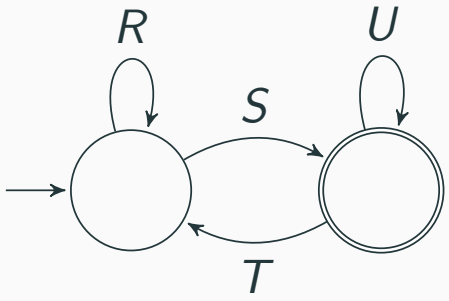
After s is eliminated.



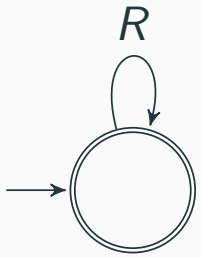
State elimination: the algorithm

For every accepting $q \in F$ eliminate all states $p \in Q \setminus \{q, q_0\}$.

- for $q \neq q_0$: $\mathrm{RegE}(q) = (R + SU^* T)^* SU^*$



- for $q = q_0$: $\mathrm{RegE}(q) = R^*$

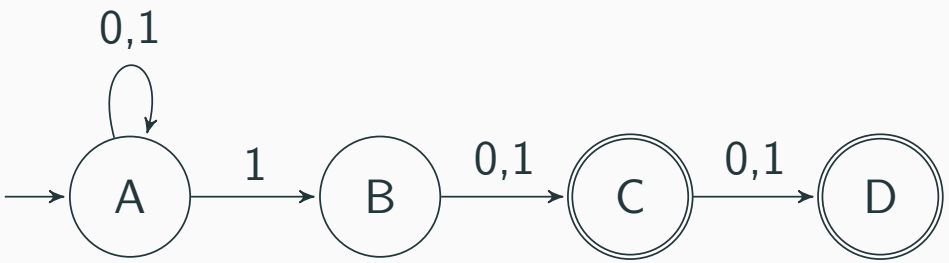


Finally, union over all accepting states: $\mathrm{RegE}(A) = \sum_{q \in F} \mathrm{RegE}(q)$

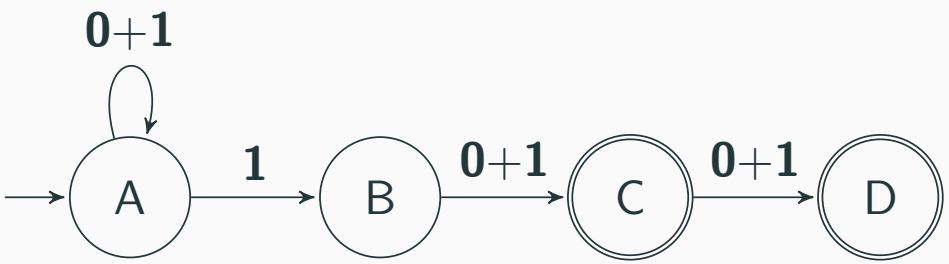
(Elimination order: first nonaccepting and noninitial states.)

State elimination: an example

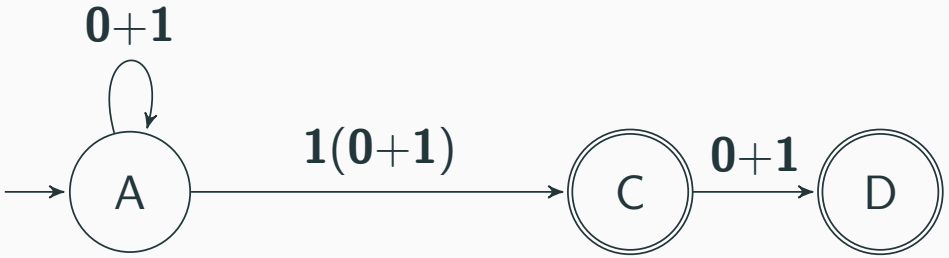
The original automaton:



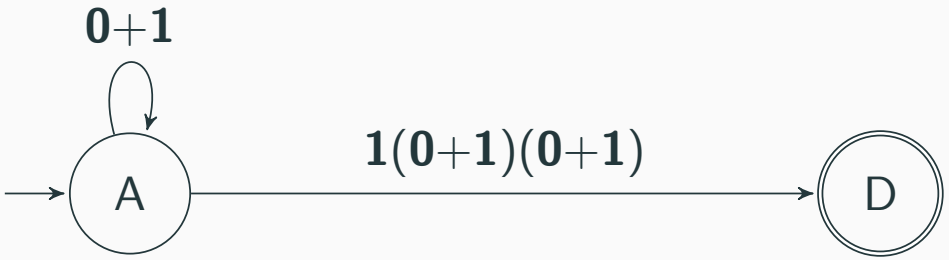
Replace letters by RE:



Eliminate B:



Eliminate C:



$$(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})+(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})(\mathbf{0}+\mathbf{1})$$

Algebraic description of regular languages

Let $\mathrm{RL}(\Sigma)$ denote the smallest set of languages over Σ that:

- contains $\emptyset$ and $\{x\}$ for any letter $x \in \Sigma$, and
- is closed under union, concatenation, and iteration.

That is, for $A, B \in \mathrm{RL}(\Sigma)$ also $A \cup B, A.B, A^* \in \mathrm{RL}(\Sigma)$. Note that:

- $\{\epsilon\} \in \mathrm{RL}(\Sigma)$ since $\{\epsilon\} = \emptyset^*$
- $\Sigma \in \mathrm{RL}(\Sigma)$ since $\Sigma = \bigcup_{x \in \Sigma} \{x\}$ (a finite union)
- $\Sigma^* \in \mathrm{RL}(\Sigma)$
- any finite language over Σ is in $\mathrm{RL}(\Sigma)$.

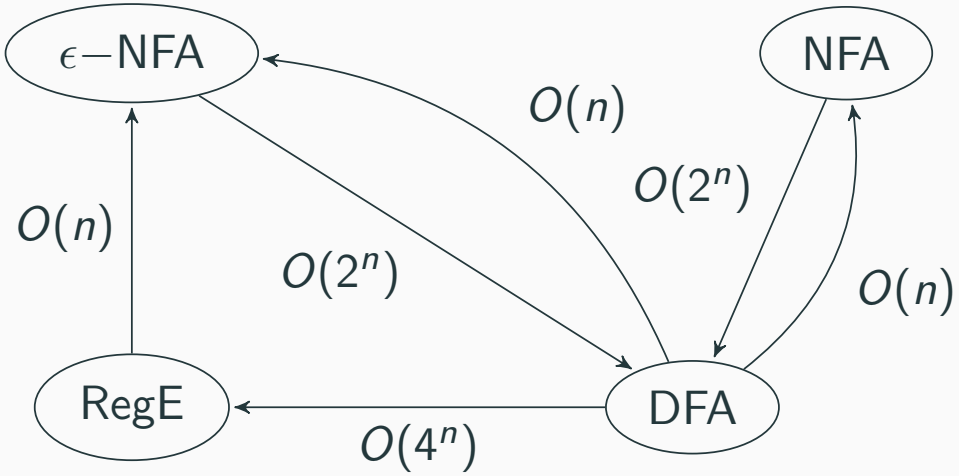
Theorem (A restatement of Kleene's Theorem)

A language over Σ is regular, iff it is in $\mathrm{RL}(\Sigma)$.

Some properties to simplify RE (will not be tested)

$$
\begin{aligned}
L.\emptyset = \emptyset.L &= \emptyset \\
\{\epsilon\}.L = L.\{\epsilon\} &= L \\
(L^*)^* &= L^* \\
(L_1 \cup L_2)^* &= L_1^*(L_2.L_1^*)^* = L_2^*(L_1.L_2^*)^* \\
(L_1.L_2)^R &= L_2^R.L_1^R \\
\partial_w(L_1 \cup L_2) &= \partial_w(L_1) \cup \partial_w(L_2) \\
\partial_w(\Sigma^* - L) &= \Sigma^* - \partial_w L.
\end{aligned}
$$

Converting between representations



- NFA or ϵ-NFA to DFA: $O(2^n)$
  - subset construction, DFA with up to 2^n states; for each state need $O(n^3)$ time to compute transitions.
  - ϵ-closure in $O(n^3)$ (search n states $\times$ n^2 arcs)
- DFA to NFA or ϵ-NFA: $O(n)$
  - a simple modification of the transition table
- DFA to RE: $O(4^n)$
- RE to ϵ-NFA: $O(n)$

1.9 String substitution

String substitution and homomorphism

A (string) substitution is a mapping $\sigma\colon \Sigma^* \to \mathcal{P}(Y^*)$ where

- Σ and Y are finite alphabets, $Y = \bigcup_{x\in\Sigma} Y_x$
- for each $x \in \Sigma$, $\sigma(x)$ is a language over Y_x
- $\sigma(\epsilon) = \{\epsilon\}$ and $\sigma(u.v) = \sigma(u).\sigma(v)$

For a language $L \subseteq \Sigma^*$, $\sigma(L) = \bigcup_{w\in L} \sigma(w) \subseteq Y^*$. A substitution is ϵ-free if no $\sigma(x)$ contains ϵ.

A (string) homomorphism is defined similarly but each letter is mapped to a single word, $h\colon \Sigma^* \to Y^*$ where $h(x) \in Y_x^*$ for $x \in \Sigma$, $h(\epsilon) = \epsilon$ and $h(u.v) = h(u).h(v)$. Then $h(L) = \{h(w) \mid w \in L\}$. It is ϵ-free if $h(x) \neq \epsilon$ for all $x \in \Sigma$.

The inverse homomorphism applied to a language $L' \subseteq Y^*$:

$$h^{-1}(L') = \{w \in \Sigma^* \mid h(w) \in L'\}$$

Examples

Example (Substitution)

- If $\sigma(0) = \{a^i b^j, i, j \geq 0\}$ and $\sigma(1) = \{cd\}$, then $\sigma(010) = \{a^i b^j cda^k b^l \mid i, j, k, l \geq 0\}$.
- $\Sigma = \{f, l, s, c, d\}$, $L = L((fsl)(cfsl)^* d)$ where
  - $\sigma(f)$ is a dictionary of first names
  - $\sigma(l)$ are last names
  - $\sigma(s) = \{`\ '\}$ (space), $\sigma(c) = \{`,'\}$, $\sigma(d) = \{`.'\}$
- A document template with symbols to be replaced by fields of database entries.

Example (Homomorphism)

- Define $h(0) = ab$ and $h(1) = \epsilon$. Then $h(0011) = abab$ and for $L = \mathbf{10^*1}$ we have $h(L) = L((ab)^*)$.
- Replace special symbols with TEX code (e.g. $h(\mu) = \backslash mu$).

Preserving regularity

Theorem

Let $L \subseteq \Sigma^*$ be regular, $h\colon \Sigma^* \to Y^*$ a homomorphism, and $\sigma\colon \Sigma^* \to \mathcal{P}(Y^*)$ a substitution.

- *The language $h(L)$ is regular.*
- *If $\sigma(x)$ is regular for all $x \in \Sigma$, then $\sigma(L)$ is also regular.*

Moreover, if $L' \subseteq Y^*$ is regular, then $h^{-1}(L')$ is also regular.

Proof for homomorphism and substitution

Homomorphism $\leftrightsquigarrow$ substitution with $\sigma(x)$ one-element (regular).

Structural induction on a RE α such that $L = L(\alpha)$.

- **Induction base:** $\emptyset$, ϵ, **a** ... easy
- **Induction step:**

$$\begin{aligned} \sigma(L(\alpha+\beta)) &= \sigma(L(\alpha)) \cup \sigma(L(\beta)) \\ \sigma(L(\alpha\beta)) &= \sigma(L(\alpha)).\sigma(L(\beta)) \end{aligned}$$

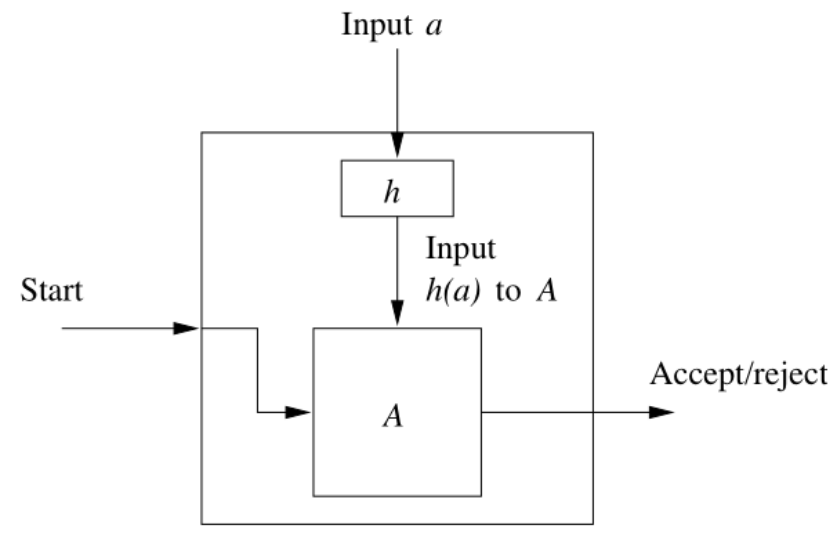
For iteration, decompose into an infinite union of powers:

$$\begin{aligned} \sigma(L(\alpha)^*) &= \sigma(L(\alpha)^0) \cup \sigma(L(\alpha)^1) \cup \ldots \\ &= \sigma(L(\alpha))^0 \cup \sigma(L(\alpha))^1 \cup \ldots = \sigma(L(\alpha))^* \end{aligned}$$

(Alternative view: take the tree of the RE α and replace every leaf x with a tree for a RE for $\sigma(x)$.) $\square$

Proof for inverse homomorphism & an example

Given a DFA $A = (Q, Y, \delta, q_0, F)$ recognizing L', construct a DFA recognizing $h^{-1}(L')$: $B(Q, \Sigma, \delta_B, q_0, F)$ with $\delta_B(q, a) = \delta^*(q, h(a))$
That is, for a letter $a \in \Sigma$ do what A does for the word $h(a)$. Easy to show (by induction on $|w|$) that $\delta_B^*(q_0, w) = \delta^*(q_0, h(w))$.



□

Example (Inverse homomorphism)

$L' = L((\mathbf{00} + \mathbf{1})^*)$, $h(a) = 01$, and $h(b) = 10$: $h^{-1}(L') = (\mathbf{ba})^*$.
[$h(L((\mathbf{ba})^*)) \in \mathbf{L}'$ is obvious, other words generate an isolated 0.]

1.10 Decision properties of regular languages

Testing emptiness

Given a representation of a regular language L, is $L = \emptyset$?

FA: is any final state reachable from the initial state? $O(n^2)$

RE: convert to $\epsilon-$NFA (in $O(n)$ time) and check reachability

or directly:

basis: $\emptyset$ is empty, ϵ and **a** are not

induction:

- $\alpha = (\alpha_1 + \alpha_2)$: empty iff both $L(\alpha_1)$ and $L(\alpha_2)$ are empty
- $\alpha = (\alpha_1\alpha_2)$ empty iff either $L(\alpha_1)$ or $L(\alpha_2)$ is empty
- $\alpha = (\alpha_1^*)$ never empty, includes ϵ

Testing membership

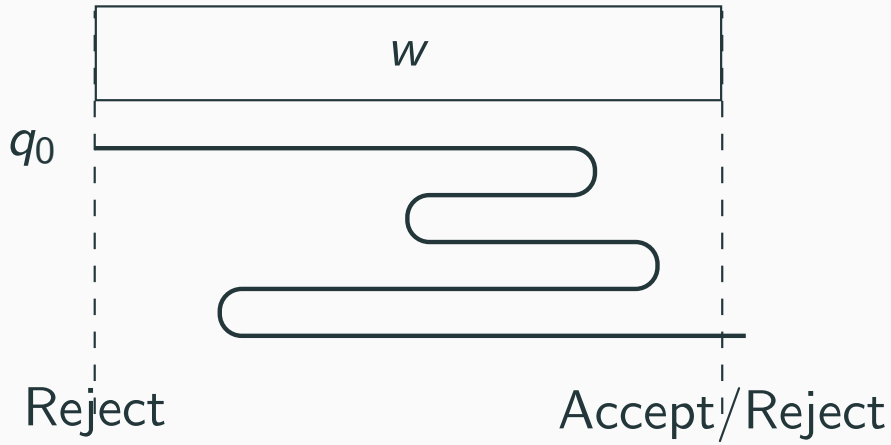
Given a regular language L and a word w, is $w \in L$?

- DFA: run the automaton; if $|w| = n$, with a suitable representation (constant time transitions) it is in $O(n)$
- NFA with s states: running time $O(ns^2)$, each letter processed by taking the previous set of states
- ϵ-NFA: first compute the ϵ-closure, then for each letter, process it and compute the ϵ-closure of the result
- RE of size s: convert to an ϵ-NFA with at most $2s$ states and then simulate, $O(ns^2)$

1.11 Two-way finite automata

Revisit previous input?

- Finite automaton makes following actions:
  - read a symbol
  - changes its state
  - moves its reading head to the right
- The head is not allowed to move to the left.
- What happens, if we allow the head to move left and right?
- **The automaton does not write anything on the tape!**



Two-way finite automaton

Definition

A Two-way DFA is a five-tuple $A = (Q, \Sigma, \delta, q_0, F)$, where Q, Σ, q_0, F are as in the definition of a DFA, and the transition function is a mapping

$$\delta : Q \times \Sigma \to Q \times \{-1, +1\}.$$

Similarly, a Two-way NFA is defined as an NFA with

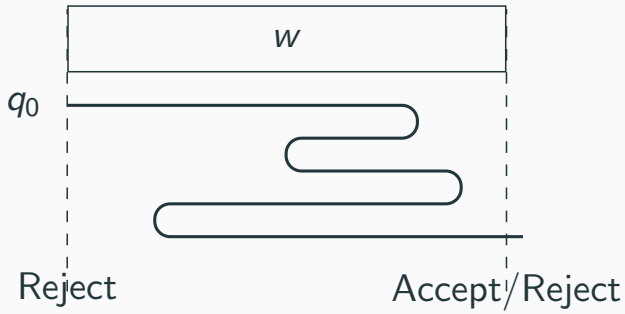
$$\delta : Q \times \Sigma \to \mathcal{P}(Q \times \{-1, +1\}).$$

- the extra value indicates direction of head movement: -1 for left, $+1$ for right
- we could allow for no movement, i.e., $\{-1, 0, +1\}$

Two-way DFA computation

A word w is accepted by a two-way DFA, if:

- computation started in q_0 at the first letter of w
- the head never moved left from the first letter of w
- the head moved right from the last letter of w
- we ended in an accepting state



NB: Infinite cycles are possible.

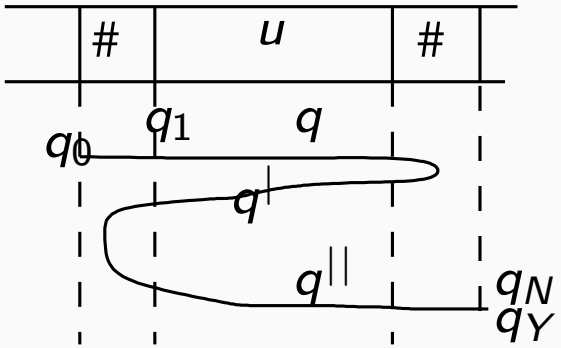
Idea: We can bookend all words with a special symbol $\# \notin \Sigma$. If $L(A) = \{\#w\# \mid w \in L\}$ is regular, so is L because $L = \partial_{\#}\partial_{\#}^{R} L(A)$

Example

Let $A = (Q, \Sigma, \delta, q_1, F)$. Define a 2-way DFA B recognizing $L(B) = \{\#u\# \mid uu \in L(A)\}$ (this is not left/right quotient!)

$$B = (Q \cup Q^{|} \cup Q^{||} \cup \{q_0, q_N, q_F\}, \Sigma, \delta^{|}, q_0, \{q_F\})$$

$\delta^{\vert}$	$x \in \Sigma$	$\#$	remark
q_0	$q_N, -1$	$q_1, +1$	q_1 is starting in A
q	$p, +1$	$q^{\vert}, -1$	$p = \delta(q, x)$
$q^{\vert}$	$q^{\vert}, -1$	$q^{\vert\vert}, +1$	
$q^{\vert\vert}$	$p^{\vert\vert}, +1$	$q_F, +1$	$q \in F, p = \delta(q, x)$
$q^{\vert\vert}$	$p^{\vert\vert}, +1$	$q_N, +1$	$q \notin F, p = \delta(q, x)$
q_N	$q_N, +1$	$q_N, +1$	
q_F	$q_N, +1$	$q_N, +1$	



Two-way finite automata and regular Languages

Theorem

Two-way DFA recognize exactly regular languages.

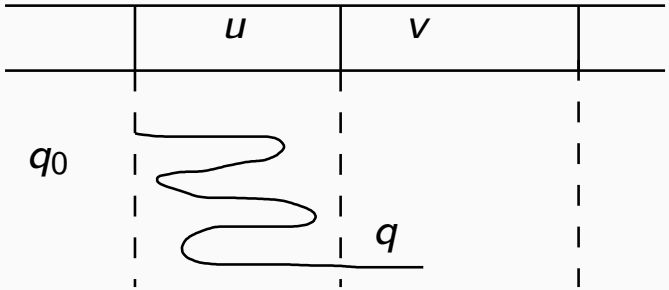
$\Leftarrow$ Convert DFA trivially to two-way DFA: $\delta'(q, a) = (\delta(q, a), +1)$

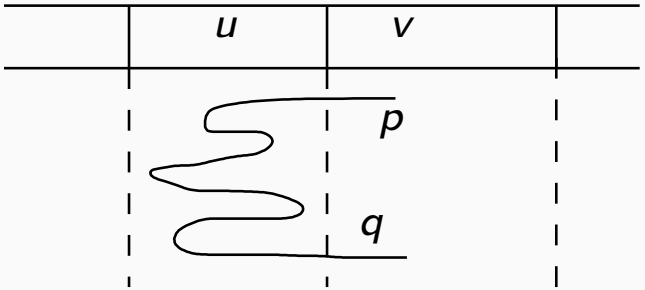
$\Rightarrow$ Use Myhill-Nerode theorem: associate with each word $u \in \Sigma^*$ a function $f_u : Q \cup \{q_0^{|}\} \to Q \cup \{\perp\}$ describing the influence of a subword u on the computation.

How exactly? See the next slide.

There are only finitely many, at most $(|Q| + 1)^{|Q|+1}$ such functions. Define $u \sim v$ iff $f_u = f_v$, verify that it is a right congruence and $L(A)$ is a union of classes. $\square$

How $f_u : Q \cup \{q_0^{|}\} \to Q \cup \{\perp\}$ describes computation over u





$f_u(q_0^{|})$ = state after first transition right in case we begin left in state q_0

$f_u(p)$ = state of right transition in case we begin right in state p

$\perp$ denotes failure (a cycle or escape left)

Summary of finite automata

- Finite Automata: DFA, reduced DFA, NFA, ϵ-NFA
- Regular Expressions
- Regular languages: closed under set operations, string operations, substitution, homomorphism, inverse hom.
- all FA and RE describe the same class of languages
- Key theorems:
  - Myhill–Nerode (implicit DFA via congruences on words)
  - Kleene (regular languages iff matched by RE)
  - Pumping lemma
- Two-way automata
- (optional) Automata with output

Summary of Lecture 4

- regular expressions
- Kleene's theorem (two variants)
- constructions: RE to ϵ-NFA, DFA to RE
- state elimination algorithm
- string substitution, homomorphism, inverse homomorphism
- decision properties
- two-way finite automata

Appendix 1: Visit every state

Visit every state

Example (visit every state)

Given a DFA A, let L consist of all $w \in \Sigma^*$ that are accepted and, moreover, during the computation every state is visited, i.e.:

- $\delta^*(q_0, w) \in F$
- for every $q \in Q$ there is a prefix x_q of w s.t. $\delta^*(q_0, x_q) = q$

We will show that this language is regular.

Construct L from $M = L(A)$ using operations preserving regularity:

- define an alphabet of 'transitions': $T = \{[paq] \mid \delta(p, a) = q\}$
- define a homomorphism $h([paq]) = a$ for all p, q, a
- $L_1 = h^{-1}(M)$ is regular (inverse homomorphism), consists of accepting sequences of transitions

Visit every state: proof continues

- start at q_0: $L_2 = L_1 \cap L(E_1.T^*)$, $E_1 = \{[q_0aq] \mid a \in \Sigma, q \in Q\}$
- adjacent states must be equal: define non-matching pairs

$$E_2 = \{[paq][rbs] \mid q \neq r, p, q, r, s \in Q, a, b \in \Sigma\}$$

  and set $L_3 = L_2 - L(T^*.E_2.T^*)$ (remove if at least one non-matching pair of adjacent states)
- L_3 already ends in accepting state: we started from $M = L(A)$
- all states: for $q \in Q$ let E_q be the RE that is the sum of all the symbols in T not containing q, set $L_4 = L_3 - \bigcup_{q \in Q}\{L(E_q^*)\}$
- from a sequence of transitions back to the word: $L = h(L_4)$

$\square$

Appendix 2: Automata with output

Automata with the output

Definition (Moore machine)

Moore machine is a sixtuple $A = (Q, \Sigma, Y, \delta, \mu, q_0)$ where

Q non–empty set of states

Σ finite nonempty set of symbols (input alphabet)

Y finite nonempty set of symbols (output alphabet)

δ a mapping $Q \times \Sigma \to Q$ (transition function)

μ a mapping $Q \to Y$ (output function)

$q_0 \in Q$ (initial state)

- the output function may imitate final states
  - $F \subseteq Q$ may be replaced by output function $\mu : Q \to \{0, 1\}$ as follows:

    $\mu(q) = 0$ if $q \notin F$,
    $\mu(q) = 1$ if $q \in F$.

Moore Machine Example

Example (Tennis Game Score)

A machine calculates the tenis score.

- Input alphabet: ID of the player who scored a point
- Output alphabet & states: the score ($Q = Y$ and $\mu(q) = q$)

State/output	A	B
00:00	15:00	00:15
15:00	30:00	15:15
15:15	30:15	15:30
00:15	15:15	00:30
30:00	40:00	30:15
30:15	40:15	30:30
30:30	40:30	30:40
15:30	30:30	15:40
00:30	15:30	00:40
40:00	A	40:15
40:15	A	40:30
40:30	A	deuce
30:40	deuce	B
15:40	30:40	B
00:40	15:00	B
deuce	A:40	40:B
A:40	A	deuce
40:B	deuce	B

Mealy machine

Definition (Mealy machine)

Mealy machine is a six–tuple $A = (Q, \Sigma, Y, \delta, \lambda_M, q_0)$ where

Q non–empty set of states

Σ finite nonempty set of symbols (input alphabet)

Y finite nonempty set of symbols (output alphabet)

δ a mapping $Q \times \Sigma \to Q$ (transition function)

λ_M a mapping $Q \times \Sigma \to Y$ (output function)

$q_0 \in Q$ (initial state)

- The output is determined by a state and the input symbol
  - Mealy machine is more general then Moore
  - The output function may be replaced as follows

$$\forall x \in \Sigma \ \lambda_M(q, x) = \mu(q)$$

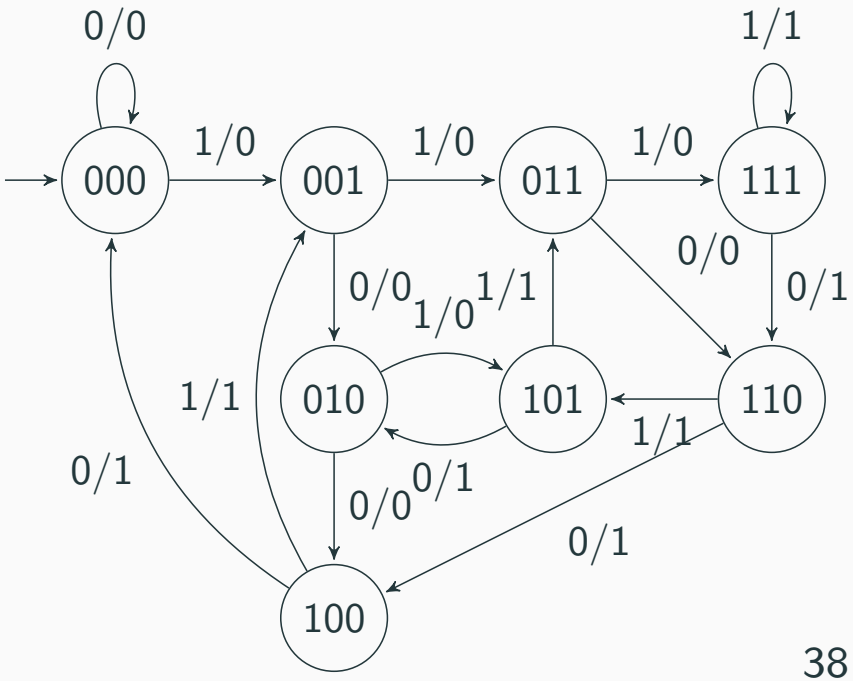
$$\text{or} \ \forall x \in \Sigma \ \lambda_M(q, x) = \mu(\delta(q, x))$$

Mealy Machine Example

The automaton for integer division of the input by 8 (the remainder is discarded).

- Three bit move to the left
- we need to remember last three bits
- three–bit dynamic memory.

State\symbol	0	1
→000	000/0	001/0
001	010/0	011/0
010	100/0	101/0
011	110/0	111/0
100	000/1	001/1
101	010/1	011/1
110	100/1	101/1
111	110/1	111/1



Extended Output Function

for any word in the input alphabet $\Sigma^* \rightarrow$ we get a word in the output alphabet Y^*

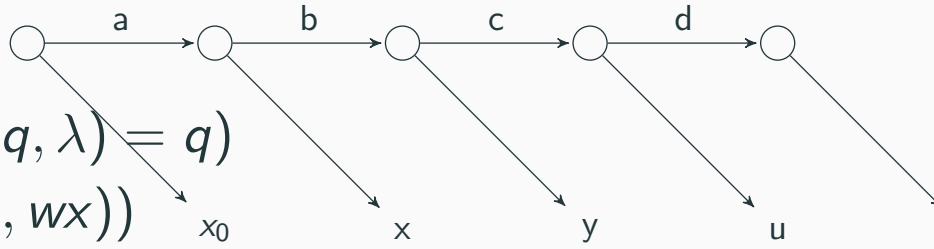
Moore machine

output function $\mu : Q \rightarrow Y$

$\mu^* : Q \times \Sigma^* \rightarrow Y^*$

$\mu^*(q, \lambda) = \lambda$ (sometimes $\mu^*(q, \lambda) = q$)

$\mu^*(q, wx) = \mu^*(q, w).\mu(\delta^*(q, wx))$



Example: μ^*(00:00,AABA)=(00:00 .) 15:00 . 30:00 . 30:15 . 40:15

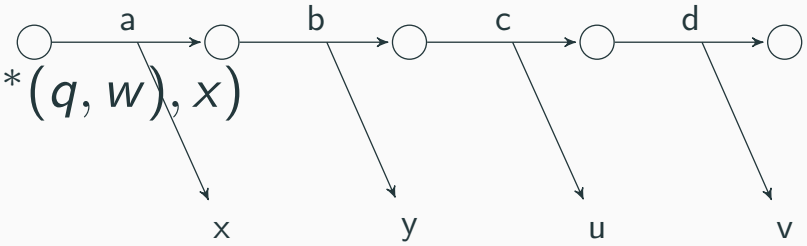
Mealy machine

output function $\lambda_M : Q \times \Sigma \rightarrow Y$

$\lambda_M^* : Q \times \Sigma^* \rightarrow Y^*$

$\lambda_M^*(q, \lambda) = \lambda$

$\lambda_M^*(q, wx) = \lambda_M^*(q, w).\lambda_M(\delta^*(q, w), x)$



Example: λ_M^*(000,1101010)=0001101

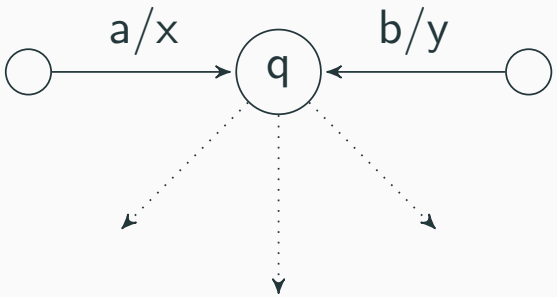
Moore and Mealy Machines Reductions

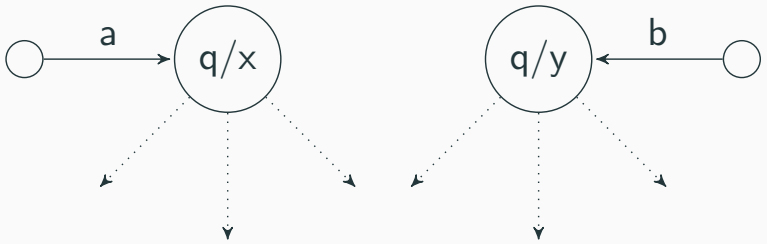
Lemma

For any Moore machine there exists a Mealy machine, and vice versa, mapping each input word to the same output word.

$\Rightarrow$ Mealy machine $B = (Q, \Sigma, Y, \delta, \lambda_M, q_0)$ where $\lambda_M(q, x) = \mu(\delta(q, x))$

$\Leftarrow$ We define states of the Moore machine $Q \times Y$, $\delta^{|}([q, y], x) = [\delta(q, x), \lambda(q, x)]), \mu([q, y]) = y$.

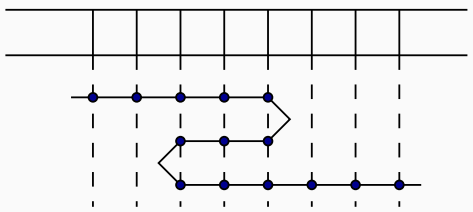




Appendix 3: A constructive conversion from two-way DFA to NFA

Constructive proof of two-way automata regularity: two-way DFA to NFA

Transcribe left-right movement to a linear computation.
We are interested in accepting computations only.
We focus on transitions in cuts between input symbols.



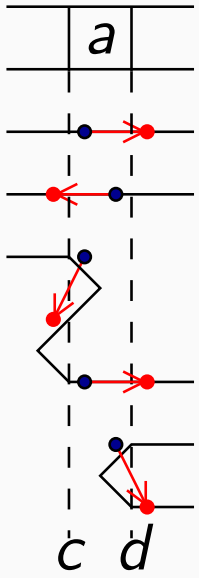
The direction of movement repeats (right, left)
The first and the last transitions are to the right
A is deterministic, accepting computations have no cycles.
The first and the last cut contain only one state.

- Find all possible cuts = state sequences (finite number).
- Define non-deterministic transition between cuts according to input symbol.
- We re–construct the computation by composing cuts like a puzzle.

Formal reduction two-way DFA to NFA

Let $A = (Q, \Sigma, \delta, q_0, F)$be a two–way DFA. We define an equivalent NFA $B = (Q^|, \Sigma, \delta^|, (q_0), F^|)$ as follows:
$Q^|$ all possible correct transition sequences

- sequences of states $(q^1, \ldots, q^k); q^i \in Q$
- with an odd length $(k = 2m + 1)$
- no state repeats at odd nor at even position
  $(\forall i \neq j)\ (q^{2i} \neq q^{2j})\&(\forall i \neq j)\ (q^{2i+1} \neq q^{2j+1})$



$F^| = \{(q)|q \in F\}$ sequences of the length 1
$\delta^|(c, a) = \{d|d \in Q^|\&c \xrightarrow{a} d$ is a locally consistent transition for $a\}$

- $\exists$ bijection: $h : c_{odd} \cup d_{even} \to c_{even} \cup d_{odd}$ so that:
- for $h(q) \in c_{even}$ is $(h(q), -1) = \delta(q, a)$
- for $h(q) \in d_{odd}$ is $(h(q), +1) = \delta(q, a)$

Trajectory in A corresponds to cuts in $B \Rightarrow L(A) = L(B)$

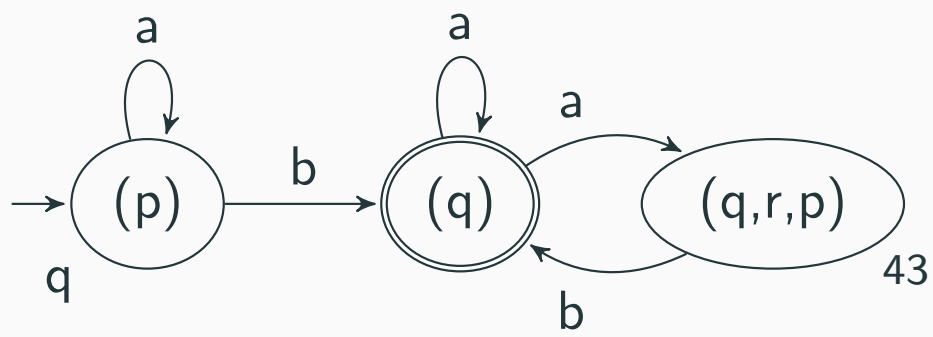
Example Reduction Two-way DFA to NFA

	a	b
$\rightarrow p$	p,+1	q,+1
$*q$	q,+1	r,-1
r	p,+1	r,-1

Possible cuts and their transitions

- leftwards only r – all even positions r, that means only one even position
- possible cuts: $(p), (q), (p, r, q), (q, r, p)$.

	a	b
$\rightarrow (p)$	(p)	(q)
$*(q)$	(q),(q,r,p)	
(p,r,q)		
(q,r,p)		(q)

Non-accepting example:

a	a	b	a	a	b	a	a	b	b
p	p	p	q	q	q				
					r				
					p	q	q	q	
								r	
								p	q